# 3. Working with Gradle: Setting Up a Gradle Project, Understanding Build Scripts (Groovy and Kotlin DSL), Dependency Management and Task Automation.

## Working with Gradle Project (Groovy DSL):

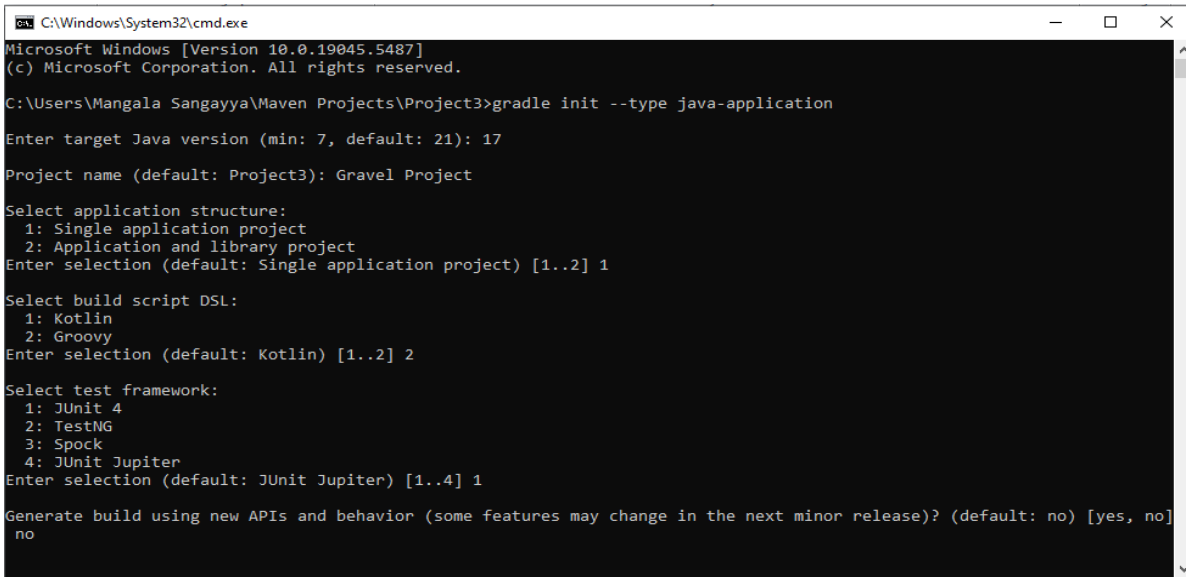
### 1. Create a new Project

Open command prompt and navigate to the created folder and give the following command to create the gradle project

***gradle init --type java-application***

Then it will ask some requirements:

- **Enter target Java version (min: 7, default: 21):** 17
- **Project name (default: program3-groovy):** groovyProject
- **Select application structure:**
  - 1: Single application project
  - 2: Application and library project
    - **Enter selection (default: Single application project) [1..2]** 1
- **Select build script DSL:**
  - 1: Kotlin
  - 2: Groovy
    - **Enter selection (default: Kotlin) [1..2]** 2
- **Select test framework:**
  - 1: JUnit 4
  - 2: TestNG
  - 3: Spock
  - 4: JUnit Jupiter
    - **Enter selection (default: JUnit Jupiter) [1..4]** 1
- **Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no]**
  - no



## 2. build.gradle (Groovy DSL):

Open the build.gradle file ***(projectFolder/app/build)*** and add the following code

```
plugins {
    // Apply the application plugin to add support for building a CLI application in Java.
    id 'application'
}

repositories {
    // Use Maven Central for resolving dependencies.
    mavenCentral()
}

dependencies {
    // Use JUnit test framework.
    testImplementation 'junit:junit:4.13.2'
}

application {
    // Define the main class for the application.
    mainClass = ‘org.example.App'
}
test {
    outputs.upToDateWhen { false }

    testLogging {
        events "passed", "failed", "skipped"
        exceptionFormat "full"
        showStandardStreams = true
    }
}
```

## 3. App.java

- Open **src/main/java/org/example/**
- Write the below code and save.

```
package org.example;
public class App {
    public static void main(String[] args) {
        double num1 = 5;
        double num2 = 10;
        double sum = num1 + num2;
        System.out.printf("The sum of %.2f and %.2f is %.2f%n", num1, num2, sum);
    }
}
```

## 4. AppTest.java (JUnit Test)

- Open **src/test/java/org/example/**
- Write the below code and save

```
package org.example;
import org.junit.Test;
import static org.junit.Assert.*;
public class AppTest {
    @Test
    public void testAddition() {
        double num1 = 5;
        double num2 = 10;
        double expectedSum = num1 + num2;
        double actualSum = num1 + num2;
        assertEquals(expectedSum, actualSum, 0.01);
    }
}
```

## 5. Run Gradle Commands:

To build the project execute below command

*gradle build*

```
C:\Users\Mangala Sangayya\Maven Projects\Project3\app>gradle build
Starting a Gradle Daemon (subsequent builds will be faster)
Calculating task graph as no cached configuration is available for tasks: build

> Task :app:test

AdditionOperationTest > testAddition PASSED

[Incubating] Problems report is available at: file:///C:/Users/Mangala%20Sangayya/Maven%20Projects/Project3/build/report
s/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own sc
ripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.13/userguide/command_line_interface.html#sec:command_line_wa
rnings in the Gradle documentation.

BUILD SUCCESSFUL in 10s
7 actionable tasks: 6 executed, 1 up-to-date
Configuration cache entry stored.
C:\Users\Mangala Sangayya\Maven Projects\Project3\app>
```

To run the project execute below command

*gradle run*

To test the project execute below command

*gradle test*

```
C:\Users\Mangala Sangayya\Maven Projects\Project3\app>gradle run
Calculating task graph as no cached configuration is available for tasks: run

> Task :app:run
The sum of 5.00 and 10.00 is 15.00

[Incubating] Problems report is available at: file:///C:/Users/Mangala%20Sangayya/Maven%20Projects/Project3/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.13/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 1s
2 actionable tasks: 1 executed, 1 up-to-date
Configuration cache entry stored.
C:\Users\Mangala Sangayya\Maven Projects\Project3\app>gradle test
Calculating task graph as no cached configuration is available for tasks: test

> Task :app:test

AdditionOperationTest > testAddition PASSED

[Incubating] Problems report is available at: file:///C:/Users/Mangala%20Sangayya/Maven%20Projects/Project3/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.13/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 1s
3 actionable tasks: 1 executed, 2 up-to-date
Configuration cache entry stored.
C:\Users\Mangala Sangayya\Maven Projects\Project3\app>
```

## Working with Gradle Project (Kotlin DSL):

### 1. Create a new Project

Open command prompt and navigate to the created folder and give the following command to create the gradle project

***gradle init --type java-application***

Then it will ask some requirements:

- **Enter target Java version (min: 7, default: 21):** 17
- **Project name (default: program3-groovy):** KotlinProject
- **Select application structure:**
  - 1: Single application project
  - 2: Application and library project
    - **Enter selection (default: Single application project) [1..2]** 1
- **Select build script DSL:**
  - 1: Kotlin
  - 2: Groovy
    - **Enter selection (default: Kotlin) [1..2]** 1
- **Select test framework:**
  - 1: JUnit 4
  - 2: TestNG
  - 3: Spock
  - 4: JUnit Jupiter
    - **Enter selection (default: JUnit Jupiter) [1..4]** 1
- **Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no]**
  - No

```
C:\Users\Mangala Sangayya\Maven Projects\Kotlin>gradle init --type java-application


Enter target Java version (min: 7, default: 21):
Project name (default: Kotlin): 17

Select application structure:
  1: Single application project
  2: Application and library project
Enter selection (default: Single application project) [1..2] 1

Select build script DSL:
  1: Kotlin
  2: Groovy
Enter selection (default: Kotlin) [1..2] 1

Select test framework:
  1: JUnit 4
  2: TestNG
  3: Spock
  4: JUnit Jupiter
Enter selection (default: JUnit Jupiter) [1..4] 1

Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no] no


> Task :init
Learn more about Gradle by exploring our Samples at https://docs.gradle.org/8.13/samples/sample_building_java_applications.html

BUILD SUCCESSFUL in 46s
1 actionable task: 1 executed
C:\Users\Mangala Sangayya\Maven Projects\Kotlin>
```

## 2. **build.gradle.kts (Kotlin DSL)**

Open the build.gradle file ***(projectFolder/app/build)*** and add the following code

```
plugins {
    kotlin("jvm") version "1.8.21"
    application
}

repositories {
    mavenCentral()
}

dependencies {
    implementation(kotlin("stdlib"))
    testImplementation("junit:junit:4.13.2")
}

application {
    mainClass.set("org.example.MainKt")
}

tasks.test {
    useJUnit()

    testLogging {
        events("passed", "failed", "skipped")
        exceptionFormat = org.gradle.api.tasks.testing.logging.TestExceptionFormat.FULL
        showStandardStreams = true
```

```
    }
    outputs.upToDateWhen { false }
}
java {
    toolchain {
        languageVersion.set(JavaLanguageVersion.of(17))
    }
}
```

3. **Main.kt** (Change the file name in **src/main/java/org/example/** and write the below code)
   - Change the file name **App.java** to **Main.kt**: use the command ***ren App.java Main.kt***

```
package org.example

fun addNumbers(num1: Double, num2: Double): Double {
    return num1 + num2
}

fun main() {
    val num1 = 10.0
    val num2 = 5.0
    val result = addNumbers(num1, num2)
    println("The sum of $num1 and $num2 is: $result")
}
```

4. **MainTest.kt** (Change the file name in **src/test/java/org/example/** and write the below code)
   - Change the file name **AppTest.java** to **MainTest.kt**: use the command ***ren AppTest.java MainTest.kt***

```
package org.example
import org.junit.Assert.*
import org.junit.Test

class MainTest {

    @Test
    fun testAddNumbers() {
        val num1 = 10.0
        val num2 = 5.0

        val result = addNumbers(num1, num2)

        assertEquals("The sum of $num1 and $num2 should be 15.0", 15.0, result, 0.001)
    }
}
```

## 5. Run Gradle Commands:

To build the project execute below command

*gradle build*

```
C:\Users\Mangala Sangayya\Maven Projects\Kotlin>gradle build
Reusing configuration cache.

> Task :app:test

MainTest > testAddNumbers PASSED

BUILD SUCCESSFUL in 2s
7 actionable tasks: 2 executed, 5 up-to-date
Configuration cache entry reused.
C:\Users\Mangala Sangayya\Maven Projects\Kotlin>gradle run
Calculating task graph as no cached configuration is available for tasks: run
```

To run the project execute below command

*gradle run*

To test the project execute below command

*gradle test*

```
C:\Users\Mangala Sangayya\Maven Projects\Kotlin>gradle run
Calculating task graph as configuration cache cannot be reused because file 'app\build.gradle.kts' has changed.

> Task :app:run
The sum of 10.0 and 5.0 is: 15.0

[Incubating] Problems report is available at: file:///C:/Users/Mangala%20Sangayya/Maven%20Projects/Kotlin/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.13/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 1s
2 actionable tasks: 1 executed, 1 up-to-date
Configuration cache entry stored.
C:\Users\Mangala Sangayya\Maven Projects\Kotlin>gradletest
'gradletest' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Mangala Sangayya\Maven Projects\Kotlin>gradle test
Calculating task graph as no cached configuration is available for tasks: test

> Task :app:test

MainTest > testAddNumbers PASSED

[Incubating] Problems report is available at: file:///C:/Users/Mangala%20Sangayya/Maven%20Projects/Kotlin/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.13/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 1s
3 actionable tasks: 1 executed, 2 up-to-date
Configuration cache entry stored.
```